Random Gradient-Free Minimization of Convex Functions

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ECE 490 Final Project

Presenter	Main responsibility
Zhihao Jin	smoothing, oracle theory, acceleration
Yuanxiang Gong	nonsmooth and stochastic optimization
Girish Ravi	smooth baseline, nonconvex extension, experiments, closing

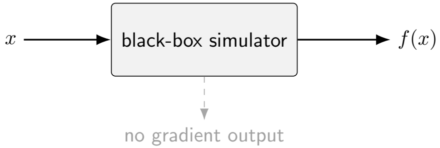
Motivation: Why Zeroth-Order Optimization?

Paper Sec. 1.1 — Owner: Leader

$$\min_{x \in \mathbb{R}^n} f(x)$$

- Classical methods need $\nabla f(x)$.
- Zeroth-order methods only query $f(x)$.
- Useful for black-box simulations, legacy code, memory-heavy AD, or nonsmooth models.

Question: what is the price of not using gradients?



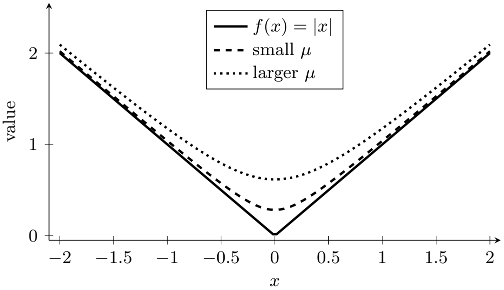
Gaussian Smoothing: The Bridge to Gradients

Paper Sec. 2 — Owner: Leader

$$f_\mu(x) = \mathbb{E}_u[f(x + \mu u)], \qquad u \sim \mathcal{N}(0, I).$$

$$\nabla f_\mu(x) = \mathbb{E}_u \left[\frac{f(x + \mu u) - f(x)}{\mu} u \right].$$

- Finite differences estimate ∇f_μ, not always ∇f.
- Nonsmooth bias: $|f_\mu(x) - f(x)| \leq \mu L_0 \sqrt{n}$.
- Smooth bias: $|f_\mu(x) - f(x)| \leq \frac{\mu^2}{2} L_1 n$.



Random Oracles and the Dimension Factor

Paper Sec. 3 — Owner: Leader

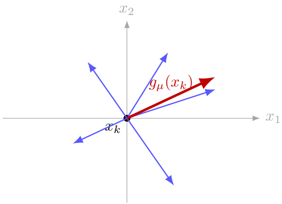
Euclidean notation shown here: $B = I$.

$$g_\mu(x) = \frac{f(x+\mu u) - f(x)}{\mu} u$$

$$\hat{g}_\mu(x) = \frac{f(x+\mu u) - f(x-\mu u)}{2\mu} u$$

$$g_0(x) = f'(x; u) u$$

$$\mathbb{E}\|g_0(x)\|^2 \le (n+4)\|\nabla f(x)\|^2$$

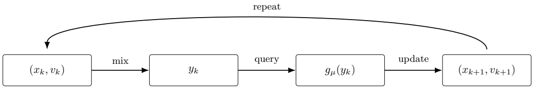


sample $u \sim \mathcal{N}(0, I)$, query finite difference, update in a random direction

- Useful direction only in expectation.
- Second moment grows with n.
- This creates the typical dimension cost.

Accelerated Random Search

Paper Sec. 6 — Owner: Leader



Nesterov-style acceleration with a random finite-difference oracle

$$\mathbb{E}[f(x_k)] - f^\star = O\left(\frac{n^2 L_1 R^2}{k^2}\right) + \text{smoothing error.}$$

- Zeroth-order analogue of Nesterov acceleration.
- Keeps the $1/k^2$ shape.
- Does not remove the dimension penalty.

Method	Full gradient	Random gradient-free
Basic smooth	$O(1/k)$	$O(n/k)$
Accelerated smooth	$O(1/k^2)$	$O(n^2/k^2)$

Nonsmooth Convex Optimization

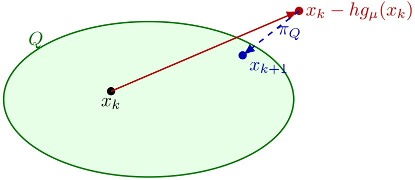
Paper Sec. 4 — Owner: Member A

$$\min_{x \in Q} f(x), \qquad f \text{ convex and nonsmooth}$$

$$x_{k+1} = \pi_Q(x_k - h_k g_\mu(x_k))$$

$$\text{For } RS_0: \quad \mathbb{E}[f(\hat{x}_N)] - f^\star = O\left(\frac{L_0 R\sqrt{n}}{\sqrt{N}}\right)$$

- Same spirit as projected subgradient descent.
- Random direction replaces a chosen subgradient.



projection returns the random step to the feasible set

Finite Differences and Smoothing Bias

Paper Sec. 4 — Owner: Member A

Method	Oracle	Message
RS_0	directional derivative	clean theory
RS_mu	forward finite difference	true zero-order
subgradient	subgradient	needs subgradient

$$f_\mu(x^\star) - f^\star \leq \mu L_0 \sqrt{n}$$

$$RS_0 : \; O(n/\varepsilon^2), \qquad RS_\mu : \; O(n^2/\varepsilon^2) \quad \text{in the nonsmooth finite-difference setting.}$$

- Smaller μ reduces smoothing bias.
- Too small μ can increase numerical cancellation.
- Theory gives safe bounds, not necessarily best empirical tuning.

Stochastic Zeroth-Order Optimization

Paper Sec. 4 — Owner: Member A

$$f(x) = \mathbb{E}_\xi[F(x, \xi)]$$

$$s_\mu(x) = \frac{F(x + \mu u, \xi) - F(x, \xi)}{\mu} u$$

$$\boxed{\text{sample } \xi} \longrightarrow \boxed{\text{sample } u} \longrightarrow \boxed{\text{same-}\xi \text{ queries}} \longrightarrow \boxed{\text{update } x}$$

Paper summary rate shape: $O\left(\frac{n}{\sqrt{k}}\right)$ in expectation.

- Randomness comes from both u and ξ.
- Reusing ξ in the difference reduces sampling noise.

Smooth Convex Baseline: Random Gradient Descent

Paper Sec. 5 — Owner: Member B

$$\min_x f(x), \qquad f \in C^{1,1}$$

$$x_{k+1} = x_k - h g_\mu(x_k), \qquad h = \frac{1}{4(n+4)L_1}$$

$$\mathbb{E}[f(\hat{x}_N)] - f^\star = O\left(\frac{nL_1R^2}{N}\right) + O(\mu^2 n^2 L_1)$$

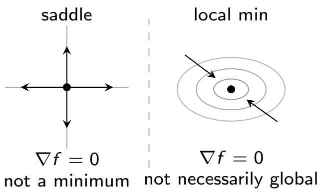
- Gradient descent: $O(1/k)$.
- Random gradient-free: still $O(1/k)$, but with a dimension factor.
- This is the theory bridge to the quadratic-chain experiment.

Nonconvex Extension: What Changes?

Paper Sec. 7 — Owner: Member B

$$\min_x f(x), \qquad f \text{ may be nonconvex}$$

Goal: make $\|\nabla f(x)\|$ small, not prove global optimality.



- Convex objective gap is no longer the right measure.
- Analyze expected gradient norm or ∇f_μ.
- The target is a stationary point.

Smooth Quadratic Experiment: Setup

Paper Sec. 8.1 — Owner: Member B

$$f_n(x) = \frac{1}{2}x_1^2 + \frac{1}{2}\sum_{i=1}^{n-1}(x_{i+1} - x_i)^2 + \frac{1}{2}x_n^2 - x_1$$

- Smooth quadratic chain from paper Sec. 8.1.
- Known optimizer, so objective gaps are computable.
- Methods: GM, RG_μ, diagnostic RG_0.
- Dimensions: $32, 64, 128$; seeds: 1:10.

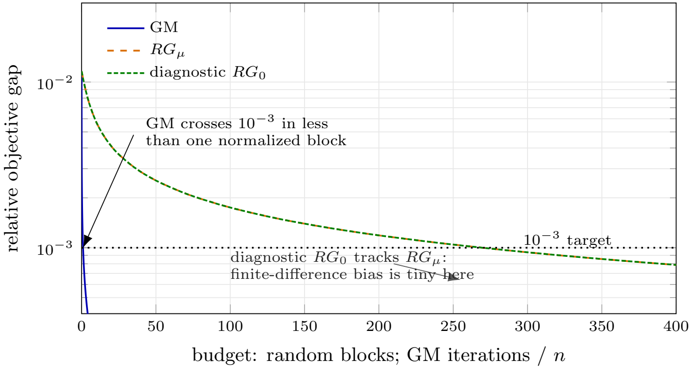
Metrics

Relative objective gap, blocks of n random directions, and function-call accounting.

$$\mu = \max\{\mu_{\mathrm{theory}}, 10^{-8}\}$$

Smooth Quadratic Experiment: Results

Paper Sec. 8.1 — Owner: Member B



Method	n	Normalized budget to 10^{-3}	Success
GM	32	3.9	100%
RG_μ	32	–	0%
GM	64	1.0	100%
RG_μ	64	270.0	100%
GM	128	0.2	100%
RG_μ	128	83.2	100%

RG rows count blocks of n random directions; GM rows report iterations divided by n.

- Full-gradient GM reaches 10^{-3} much earlier.
- RG_μ decreases steadily and nearly matches diagnostic RG_0.
- Since one RG block uses n random directions, this supports the dimension-cost story; it is not a proof.

Final Takeaways and Limitations

Paper Sec. 8.4 — Owner: Member B

- Gaussian smoothing makes finite-difference random vectors estimate ∇f_μ in expectation.
- Random gradient-free methods mimic first-order methods with extra dimension cost.
- Acceleration is possible: smooth convex rate shape $O(n^2/k^2)$.
- Use these methods when gradients are unavailable, unreliable, or too costly.

Setting	Classical method shape	Random gradient-free shape
Nonsmooth convex	$O(1/\sqrt{k})$	$O(\sqrt{n/k})$ or worse for finite differences
Smooth convex	$O(1/k)$	$O(n/k)$
Accelerated smooth	$O(1/k^2)$	$O(n^2/k^2)$
Nonconvex smooth	Stationarity	Expected gradient-norm control

Limitations

Worse high-dimensional scaling, sensitivity to μ, and slower than accurate gradient methods when gradients are available.

Backup B1: Unified Notation

Symbol	Meaning
n	dimension of the decision variable
$f(x)$	original objective function
$f_\mu(x)$	Gaussian-smoothed objective, $\mathbb{E}[f(x+\mu u)]$
u	Gaussian random direction, usually $u \sim \mathcal{N}(0, I)$
μ	smoothing or finite-difference radius
$g_\mu(x)$	forward finite-difference random oracle
$\hat{g}_\mu(x)$	symmetric finite-difference random oracle
$g_0(x)$	directional-derivative oracle
L_0, L_1	Lipschitz constants for nonsmooth and smooth cases
R	initial distance bound, typically $\|x_0 - x^\star\|$
Q, π_Q	feasible set and Euclidean projection

For presentation we use the Euclidean case $B = I$; the paper states the general geometry with Bu in the oracle and $B^{-1}g$ in the update.

Backup B2: Gaussian Gradient Identity

$$f_\mu(x) = \frac{1}{\mu^n \kappa} \int f(y) \exp\left(-\frac{\|y-x\|^2}{2\mu^2}\right)\, dy.$$

- Change variables with $y = x + \mu u$.
- Differentiate under the integral.
- Use $\mathbb{E}[u] = 0$ to subtract $f(x)$ inside the expectation.

$$\nabla f_\mu(x) = \mathbb{E}_u \left[\frac{f(x+\mu u) - f(x)}{\mu} u \right].$$

Backup B3: Accelerated Method Skeleton

1. Maintain two sequences: current point x_k and auxiliary point v_k.
2. Form an intermediate point y_k from x_k and v_k.
3. Query a random oracle $g_\mu(y_k)$.
4. Update both x_{k+1} and v_{k+1} using the oracle.

$$\mathbb{E}[f(x_k)] - f^\star = O\left(\frac{n^2 L_1 R^2}{k^2}\right) + \text{smoothing error.}$$

Presentation rule

Show the structure and rate on the main slide; keep the parameter algebra here.

Backup B4: Function Call Accounting

- g_μ usually needs $f(x)$ and $f(x + \mu u)$, unless $f(x)$ is cached.
- $\hat{g}_\mu$ needs two shifted function values.
- Paper tables often report blocks of n random iterations.
- A gradient evaluation is not universally comparable to one function call.

Backup B5: Dimension-Scaling Summary

Method	n	Normalized budget to 10^{-3}	Success
GM	32	3.9	100%
RG_μ	32	–	0%
GM	64	1.0	100%
RG_μ	64	270.0	100%
GM	128	0.2	100%
RG_μ	128	83.2	100%

- The table reports normalized budget to reach relative gap 10^{-3}.
- RG_μ rows count blocks of n random directions; GM rows report iterations divided by n.
- `GM` uses full gradients, so its cost is not directly comparable to function calls.
- RG_μ is the function-value random method; RG_0 is diagnostic and appears in the detailed CSV.